

ELI5: Why does ChatGPT use so much energy?

Posted by u/Fun_Ad_7163 • 0 points • 244 comments

Recently saw a post that ChatGPT uses more power than the entire New York city

Comments

u/grafknives • 1 points • 2025-10-10 17:34 UTC

Ok, ELI5 version.

It uses a lot of power because it makes A LOT of math.

1. LLMs generate one word(or truly part of word) at time.
2. And LLMs need to mathematically process whole input for every word it will generate in response.
3. And LLMs generate those by doing computation on absolutely insanely large data.
4. And LLMs generally don't save common responses, so they have to do it over and over again every time.

u/gmanflnj • 1 points • 2025-10-09 15:13 UTC

When they say "chat gpt" uses energy, they mean "many warehouse sized data centers all running many many thousands of graphics cards"

u/gmanflnj • 1 points • 2025-10-09 15:14 UTC

Graphics cards, because the type of math that is behind machine learning is similar to/the same as the math used to make graphics.

u/yobarisushcatel • 1 points • 2025-10-09 15:12 UTC

Takes a lot to cool them down as well as powering them, imagine how much power everyone playing video games uses

u/PepSakdoek • 1 points • 2025-10-09 09:13 UTC

Training a model takes a lot of energy, answering the questions are a lot less energy.

But they are constantly giving him incremental extra information so it is sort of constantly being trained more and more.

u/wizzard419 • 1 points • 2025-10-09 02:17 UTC

Part of that is a bit of a misdirect. ChatGPT uses a lot of energy, but eating a burger (for example) can still be more harmful.